13224917

ROBOTICS AND ARTIFICIAL INTELLIGENCE

Maximum Marks: 100

Time allowed: Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
2. *You will **not** be allowed to write during the first **15** minutes.*
3. *This time is to be spent in reading the question paper.*
4. ***The time given at the head of this Paper is the time allowed for writing the answers.***

5. *Attempt **all** questions from **Section A** and **any four** questions from **Section B.***
6. *The intended marks for questions or parts of questions are given in brackets[ ].*

Instruction for the Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.

SEAL

This paper consists of 12 printed pages.

T26 661 **Turn Over**

SECTION A (40 Marks)

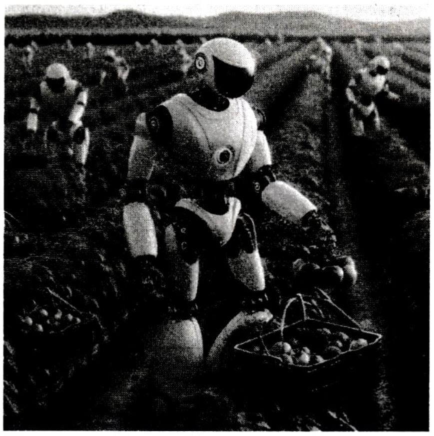
*(Attempt **all** questions from this Section.)*

Question 1

Choose the correct answers to the questions from the given options. [20]

(Do not copy the questions. Write the correct answers only.)

(i) Study the image given below.



In which field is this robot used?

(a) Defence

(b) Agriculture

(c) Medicine

(d) Entertainment

(ii) ________ is an example of *deterministic computing*.

(a) Traffic light sequencing

(b) Autonomous driving

(c) Stock market prediction

(d) Face recognition

(iii) A room heater is an example of a/an:

(a) gear

(b) robot

(c) actuator

(d) machine

(iv) Tinkercad is commonly used for:

(a) simulating robotic movements.

(b) data encryption.

(c) performing gear ratio calculations only.

(d) building operating systems.

(v) In the block diagram of a robotic control system, the correct sequence is:

(a) Input → Robot → Controller → Feedback

(b) Feedback → Input → Controller → Robot

(c) Input → Controller → Robot → Feedback

(d) Controller → Input → Robot → Feedback

(vi) Which of the following is an example of a linear actuator?

(a) DC motor

(b) Pneumatic piston

(c) Arduino

(d) Electric fan

(vii) From the following, which function of Python checks whether all characters in a string *'s'* are alpha numeric?

(a) s.allnum( )

(b) s.isnumal( )

(c) s.isalnum( )

(d) s.isalphanum( )

(viii) A motor drives a small gear with 10 teeth which is connected to a larger gear with 40 teeth. What is the gear ratio?

(a) 4 : 1

(b) 1 : 4

(c) 1 : 2

(d) 2 : 1

(ix) Which of the following string operators in Python is used to find the *length* of a string *'a'*?

(a) len(a)

(b) len()

(c) a.len()

(d) a.len(a)

(x) Unauthorised hardware access leads to:

(a) loss of data.

(b) increased computer speed.

(c) better performance.

(d) having more data.

(xi) In machine learning, *testing data* is primarily used to:

(a) train the model for better performance.

(b) measure model performance on unseen data.

(c) store training parameters.

(d) provide labels for training.

(xii) The function of a sensor is to:

(a) detect changes in environment and send signals.

(b) process the input data and make decisions.

(c) change electrical energy into motion.

(d) supply power to the robot.

(xiii) The purpose of *Turing Test* is to evaluate:

(a) memory consumption of a computer.

(b) the processing speed of computers and humans.

(c) two computer programs.

(d) if a machine can exhibit human-like responses.

(xiv) __________ software is used to simulate a robotic system.

(a) LibreOffice

(b) MS Word

(c) Tinkercad

(d) MySQL

(xv) **Assertion (A):** Robots use actuators for movement.

Reason (R): Actuators sense the environment and convert it to electrical signals.

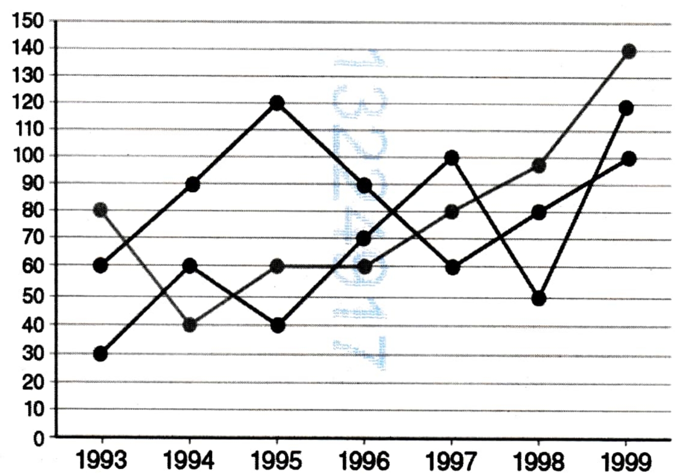
(a) (A) is true but (R) is false.

(b) (A) is false but (R) is true.

(c) Both (A) and (R) are true and (R) is the correct explanation for (A).

(d) Both (A) and (R) are true but (R) is not the correct explanation for (A).

(xvi) Identify the *type of graph* shown below:



(a) Scatter plot

(b) Bar graph

(c) Pie chart

(d) Line graph

(xvii) In AI problem scoping, "**Who**" refers to the:

(a) dataset used for training.

(b) end users affected by the project.

(c) programming environment used.

(d) hardware requirements of the project.

(xviii) Give the output of the following Python code:

```
msg = "Artificial Intelligence"
print(msg[11:14])
```

(a) tel

(b) Int

(c) ell

(d) gen

(xix) The *keyword* used to include a package or module in a Python program is:

(a) export

(b) include

(c) collect

(d) import

(xx) What will be the output of the Python code given below?

```
x="BOARD EXAMINATION"
print(x[0:4:2])
```

(a) BOAR

(b) BA

(c) OR

(d) Error

Question 2

Answer the following questions:

(i) Mention *any two* sensors used in autonomous drones. [2]

(ii) *'In a hospital, cobots assist nurses in delivering medicines to patients.'* [2]

Give *any two* advantages of using cobots in hospitals.

(iii) What is spur gear? Mention *one* application of spur gear in robotics. [2]

(iv) Explain the role of data in Machine Learning. [2]

(v) How is machine intelligence similar to human intelligence? [2]

(vi) Mention *any one* benefit of using Kaggle for data acquisition. [2]

(vii) Why is modelling important in AI project framework? [2]

(viii) What will be the output of the following Python code? [2]

```
a=5
b=3
print( a > b and b<3)
```

(ix) What will be the output of the following Python code? [2]

```
str1 = 'PYTHON'
print(str1[1:4])
```

(x) What will be the output of the following Python code? [2]

```
s="Great Day"
print(s.count('a'))
print(s.startswith('g'))
```

SECTION B (60 Marks)

*(Answer **any four** questions from this Section.)*

The answers in this section should consist of the programs in either Python environment or any program environment with Python as the base.

Each program should be written using variable description / mnemonic codes so that the logic of the program is clearly depicted.

Flowcharts and algorithms are not required.

Question 3

(i) What are *Assistant Robots*? [3]

(ii) Give *any two* differences between *deterministic computing* and *probabilistic computing* with an example. [3]

(iii) A company XYZ Ltd. records its monthly sales in lakhs as follows: [9]

Month = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

Sales = [25, 30, 28, 35, 40, 38]

Write a Python program using Matplotlib to:

1. Store the data.
2. Plot a bar chart with month on X-axis and sales on Y-axis.
3. Label the axes and give a title "XYZ Ltd. Half-Yearly Sales"

Question 4

(i) What are smart homes? Name *any two* features that make a home a smart home. [3]

(ii) Briefly explain *Data Exploration* from AI project framework. [3]

(iii) Write a Python program to plot a *pie chart* comparing marks of five students as given below: [9]

stud_name = ['Jessica', 'Peter', 'Alexa', 'James', 'Ram']

marks = [48, 22, 14, 30, 49]

Import the necessary libraries.

Question 5

(i) Name the *type of sensor* used for each of the following purposes: [3]

(a) Detecting obstacles

(b) Measuring light intensity

(c) Measuring temperature

(ii) Explain *software theft*. Mention *any two* consequences of software theft. [3]

(iii) Write a Python program to do the following tasks. [9]

1. Input a sentence and store it to a string variable.
2. Print the length of the string, its first character, and its last character.
3. Use a loop to traverse through the string and print the total number of digits present.

Question 6

(i) Briefly describe the different types of joints in robotics. [3]

(ii) What is data theft? Give *any two* causes of data theft. [3]

(iii) Write a Python program to: [9]

1. Create a list of 10 numbers.
2. Print the elements in the 2^{nd} and 4^{th} index of the list.
3. Sort the list in ascending order.
4. Ask the user to enter a number and search whether it exists in the list or not.

Question 7

(i) Explain the role of controllers in a robotic system. How do they work together with sensors and actuators? [3]

(ii) Briefly explain *supervised learning*. [3]

(iii) Write a Python program that performs the following tasks on a tuple. [9]

1. Create a tuple with elements (12, 27, 30, 47, 55, 49)
2. Convert tuple into list.
3. Remove element 30 from the list.
4. Append a new list with elements [5, 25]
5. Insert element 11 at index 3.
6. Print the list.
7. Convert the list to a tuple.

Question 8

(i) What is Tinkercad? Give *any two* features of the Tinkercad workspace. [3]

(ii) Give *any three* different ways to represent data to gather meaningful information. [3]

(iii) Write a Python program that performs the following operations on a string: [9]

1. Define a string with the value "Machine Learning".
2. Convert the string to lowercase and print it.
3. Replace the word "Machine" with "Deep".
4. Check if the string ends with "Learning" and print the result (True/False).
5. Count and print the number of occurrences of the letter "n".